

Lesson 4-3: Worked Examples

Multiplying and Dividing Rational Expressions

Example 1 — Write Equivalent Rational Expressions

Write an expression equivalent to $\frac{x+3}{x+9}$. For what domain are the expressions equivalent?

Strategy: Multiply by factors equal to 1 to produce an equivalent rational expression. Here, use $\frac{x-6}{x-6}$ and $\frac{x}{x}$.

$$\begin{aligned}\frac{x+3}{x+9} &= \frac{x+3}{x+9} \cdot 1 \cdot 1 \\ &= \frac{x+3}{x+9} \cdot \frac{x-6}{x-6} \cdot \frac{x}{x} \\ &= \frac{(x+3)(x-6)x}{(x+9)(x-6)x} \\ &= \frac{x^3 - 3x^2 - 18x}{x^3 + 3x^2 - 54x}\end{aligned}$$

Since $x^3 + 3x^2 - 54x = x(x+9)(x-6)$, the new expression is undefined at $x = 0$, $x = -9$, and $x = 6$. Those values must be excluded from the domain.

The two forms are equal on the intersection of their domains. So

$$\frac{x+3}{x+9} = \frac{x^3 - 3x^2 - 18x}{x^3 + 3x^2 - 54x} \quad \text{for } \{x \mid x \in \mathbb{R}, x \neq 0, 6, -9\}.$$

Try It!

1. Write an expression equivalent to $\frac{x-4}{x}$ over the domain $\{x \mid x \neq 0 \text{ or } -2\}$.

Example 2 — Simplify a Rational Expression

Simplify the rational expression, and state the domain on which the identity holds:

$$\frac{4 - x^2}{x^2 + 3x - 10}$$

Strategy: A rational expression is in **simplified form** when the numerator and denominator share no common factors other than 1. Factor first, then cancel.

$$\begin{aligned}\frac{4 - x^2}{x^2 + 3x - 10} &= \frac{(2 - x)(2 + x)}{(x - 2)(x + 5)} \\ &= \frac{-(x - 2)(x + 2)}{(x - 2)(x + 5)} && \text{since } 2 - x = -(x - 2) \\ &= -\frac{\cancel{(x - 2)}(x + 2)}{\cancel{(x - 2)}(x + 5)} \\ &= -\frac{x + 2}{x + 5}\end{aligned}$$

Common Error. The factor $(2 - x)$ does not look identical to $(x - 2)$, but $2 - x = -1 \cdot (x - 2)$. Always check for sign-flipped common factors before declaring two polynomials have no factor in common.

$$\frac{4 - x^2}{x^2 + 3x - 10} = -\frac{x + 2}{x + 5}, \quad x \neq 2 \text{ or } -5.$$

Try It!

2. Simplify each expression and state the domain.

a. $\frac{x^2 + 2x + 1}{x^3 - 2x^2 - 3x}$

b. $\frac{x^3 + 4x^2 - x - 4}{x^2 + 3x - 4}$

Example 3 — Multiply Rational Expressions

A. What is the product of $\frac{2xy}{z}$ and $\frac{3x^2}{4yz}$?

Multiply numerators and denominators just as you would with numerical fractions, then cancel common factors.

$$\begin{aligned}\frac{2xy}{z} \cdot \frac{3x^2}{4yz} &= \frac{(2xy)(3x^2)}{z(4yz)} \\ &= \frac{\cancel{2} \cdot 3 \cdot \cancel{y} \cdot x \cdot x^2}{\cancel{2} \cdot 2 \cdot \cancel{y} \cdot z \cdot z^2} \\ &= \frac{3x^3}{2z^2}, \quad y \neq 0, z \neq 0\end{aligned}$$

B. Find the product in simplified form:

$$\frac{5x}{x+3} \cdot \frac{x^2+x-6}{x^2+2x+1} \cdot \frac{x^2+x}{5x-10}$$

Factor everything, then identify common factors.

$$\begin{aligned}\frac{5x}{x+3} \cdot \frac{x^2+x-6}{x^2+2x+1} \cdot \frac{x^2+x}{5x-10} \\ &= \frac{5x(x+3)(x-2)x(x+1)}{(x+3)(x+1)(x+1) \cdot 5(x-2)} \\ &= \frac{\cancel{5}x\cancel{(x+3)}\cancel{(x-2)}x\cancel{(x+1)}}{\cancel{5}\cancel{(x+3)}(x+1)\cancel{(x+1)}\cancel{(x-2)}} \\ &= \frac{x^2}{x+1}\end{aligned}$$

$$\frac{5x}{x+3} \cdot \frac{x^2+x-6}{x^2+2x+1} \cdot \frac{x^2+x}{5x-10} = \frac{x^2}{x+1}, \quad x \neq -3, -1, \text{ or } 2.$$

Use Structure. The product of any two rational expressions is itself a rational expression. This is one piece of evidence that rational expressions are *closed* under multiplication.

Try It!

3. Find the simplified form of each product, and state the domain.

a. $\frac{x^2 - 16}{9 - x} \cdot \frac{x^2 + x - 90}{x^2 + 14x + 40}$

b. $\frac{x + 3}{4x} \cdot \frac{3x - 18}{6x + 18} \cdot \frac{x^2}{4x + 12}$

Example 4 — Multiply a Rational Expression by a Polynomial

Find the product of $\frac{x+2}{x^4-16}$ and x^3+4x^2-12x .

Strategy: Write the polynomial as a rational expression with denominator 1. This makes it easier to track positions of factors when canceling.

$$\begin{aligned}\frac{x+2}{x^4-16} \cdot (x^3+4x^2-12x) &= \frac{x+2}{x^4-16} \cdot \frac{x^3+4x^2-12x}{1} \\ &= \frac{(x+2)x(x^2+4x-12)}{1 \cdot (x^2+4)(x^2-4)} \\ &= \frac{(x+2)x(x+6)(x-2)}{(x^2+4)(x+2)(x-2)} \\ &= \frac{\cancel{(x+2)}x(x+6)\cancel{(x-2)}}{(x^2+4)\cancel{(x+2)}\cancel{(x-2)}} \\ &= \frac{x(x+6)}{x^2+4}\end{aligned}$$

Common Error. Only common *factors* cancel, not common *terms*. You cannot cancel an x out of $\frac{x}{x^2+4}$ — the x^2+4 in the denominator is a sum, not a product involving x .

$$\frac{x+2}{x^4-16} \cdot (x^3+4x^2-12x) = \frac{x(x+6)}{x^2+4}, \quad x \neq -2 \text{ or } 2.$$

Try It!

4. Find the simplified form of each product, and state the domain.

a. $\frac{x^3-4x}{6x^2-13x-5} \cdot (2x^3-3x^2-5x)$

b. $\frac{3x^2+6x}{x^2-49} \cdot (x^2+9x+14)$

Example 5 — Divide Rational Expressions

Find the quotient of $\frac{x^3 + 3x^2 + 3x + 1}{1 - x^2}$ and $\frac{x^2 + 5x + 4}{x^2 + 3x - 4}$.

Strategy: Division by a rational expression is multiplication by its reciprocal.

$$\begin{aligned}
 & \frac{x^3 + 3x^2 + 3x + 1}{1 - x^2} \div \frac{x^2 + 5x + 4}{x^2 + 3x - 4} \\
 &= \frac{x^3 + 3x^2 + 3x + 1}{1 - x^2} \cdot \frac{x^2 + 3x - 4}{x^2 + 5x + 4} \\
 &= \frac{(x+1)(x+1)(x+1) \cdot (x+4)(x-1)}{-(x-1)(x+1) \cdot (x+1)(x+4)} \\
 &= \frac{(x+1)}{-1} \cdot \frac{\cancel{(x+1)}}{\cancel{(x+1)}} \cdot \frac{\cancel{(x+1)}}{\cancel{(x+1)}} \cdot \frac{\cancel{(x-1)}}{\cancel{(x-1)}} \cdot \frac{\cancel{(x+4)}}{\cancel{(x+4)}} \\
 &= -(x+1)
 \end{aligned}$$

Use Structure. This problem can also be written as a complex fraction:

$$\frac{\frac{x^3 + 3x^2 + 3x + 1}{1 - x^2}}{\frac{x^2 + 5x + 4}{x^2 + 3x - 4}}$$

Any complex fraction can be rewritten as a division statement.

$$\frac{x^3 + 3x^2 + 3x + 1}{1 - x^2} \div \frac{x^2 + 5x + 4}{x^2 + 3x - 4} = -(x+1), \quad x \neq -4, -1, \text{ or } 1.$$

Try It!

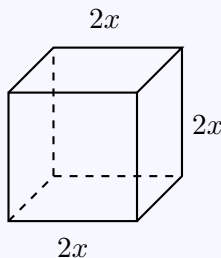
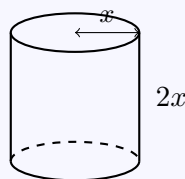
5. Find the simplified quotient and the domain of each expression.

a. $\frac{1}{x^2 + 9x} \div \frac{6 - x}{3x^2 - 18x}$

b. $\frac{2x^2 - 12x}{x + 5} \div \left(\frac{x - 6}{x + 5} \right)$

Example 6 — Use Division of Rational Expressions (Application)

A company is evaluating two packaging options. The more efficient design will have the smaller ratio of surface area to volume. Should the company use a cube-shaped box or a cylinder?

**Option 1: Cube****Option 2: Cylinder**

Setup. Let SA = surface area, V = volume. Compute the efficiency ratio $\frac{SA}{V}$ for each.

Option 1: Cube (side $2x$)

$$\begin{aligned} SA &= 2(2x)^2 + 4(2x)^2 \\ V &= (2x)^3 \end{aligned}$$

$$\begin{aligned} \frac{SA}{V} &= \frac{2(4x^2) + 4(4x^2)}{8x^3} \\ &= \frac{24x^2}{8x^3} \\ &= \frac{3}{x} \end{aligned}$$

Option 2: Cylinder (radius x , height $2x$)

$$\begin{aligned} SA &= 2\pi x^2 + 2\pi x(2x) \\ V &= \pi x^2(2x) \end{aligned}$$

$$\begin{aligned} \frac{SA}{V} &= \frac{2\pi x^2 + 4\pi x^2}{2\pi x^3} \\ &= \frac{6\pi x^2}{2\pi x^3} \\ &= \frac{3}{x} \end{aligned}$$

Comparison. Cube: $\frac{3}{x}$ Cylinder: $\frac{3}{x}$

For any positive value of x , the two designs have identical efficiency ratios. The company should choose based on other criteria (cost, manufacturing, stacking, branding, etc.).

Try It!

6. The company compares the ratios of surface area to volume for two more containers. One is a rectangular prism with a square base. The other is a rectangular prism with a rectangular base where one side equals the side length of the first container and the other side is twice as long. The surface area of the second container is $4x^2 + 6xh$. The heights of the two containers are equal. Which has the smaller surface-area-to-volume ratio?